

AI JOB REPLACEMENT

Impact of Artificial Intelligence on Jobs, Salary & Skills

A clean data-analysis presentation using R visualizations, trend interpretation, and chart callouts.

15,000

records

20

variables

2020–2026

year range

8

industries

R Programming • ggplot2 • dplyr • tidyr • corrplot • ggally

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Project Focus & Objective

What is being studied?

Automation Risk

Measures how much a job role can be affected or replaced by AI-driven automation. High-risk jobs often contain repetitive or rule-based tasks.

How AI affects work

Salary Impact

Compares salary before and after AI impact to understand whether the market rewards or reduces value for each role.

What should people learn?

Reskilling Need

Connects skill gap with urgency to learn new AI-ready skills, showing where training is most important.

The project does not simply say “AI will replace jobs”. It explains how AI changes risk, salary, industry behavior, and future skill requirements.

Dataset Snapshot

15,000

Total records

20

Dataset columns

2020–2026

Time period

8

Industries

9

Countries

Main variables used in analysis

automation_risk_percent

Risk of job automation by AI

salary_before_usd / salary_after_usd

Salary comparison before and after AI

ai_adoption_level

How strongly an industry adopts AI

skill_gap_index

Gap between current and required skills

reskilling_urgency_score

Urgency to learn new skills

ai_disruption_intensity

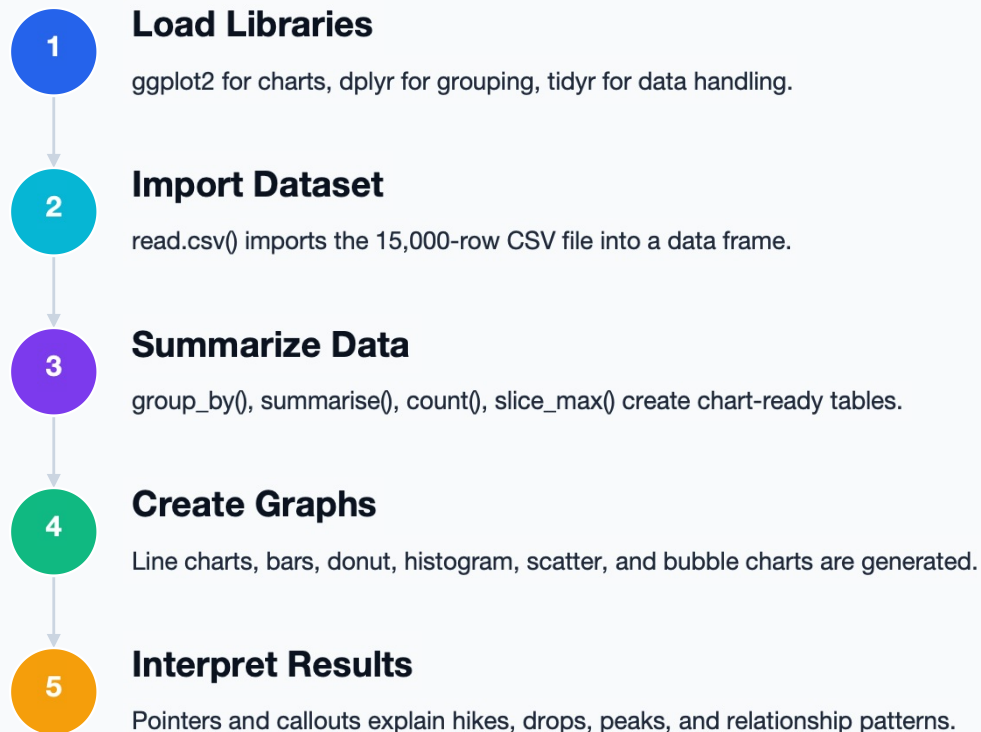
Overall intensity of AI impact

Why these variables matter

Together, these variables connect four important questions: Which jobs face AI risk? Which industries are adopting AI faster? How are salaries changing? Which skills should workers learn next?

Clean dataset structure makes it suitable for grouped summaries, trend charts, distribution plots, and relationship analysis.

Analysis Workflow in R



Key code logic

Most trend graphs use `stat_summary(fun = mean)`, which means R first calculates average values and then plots those averages.

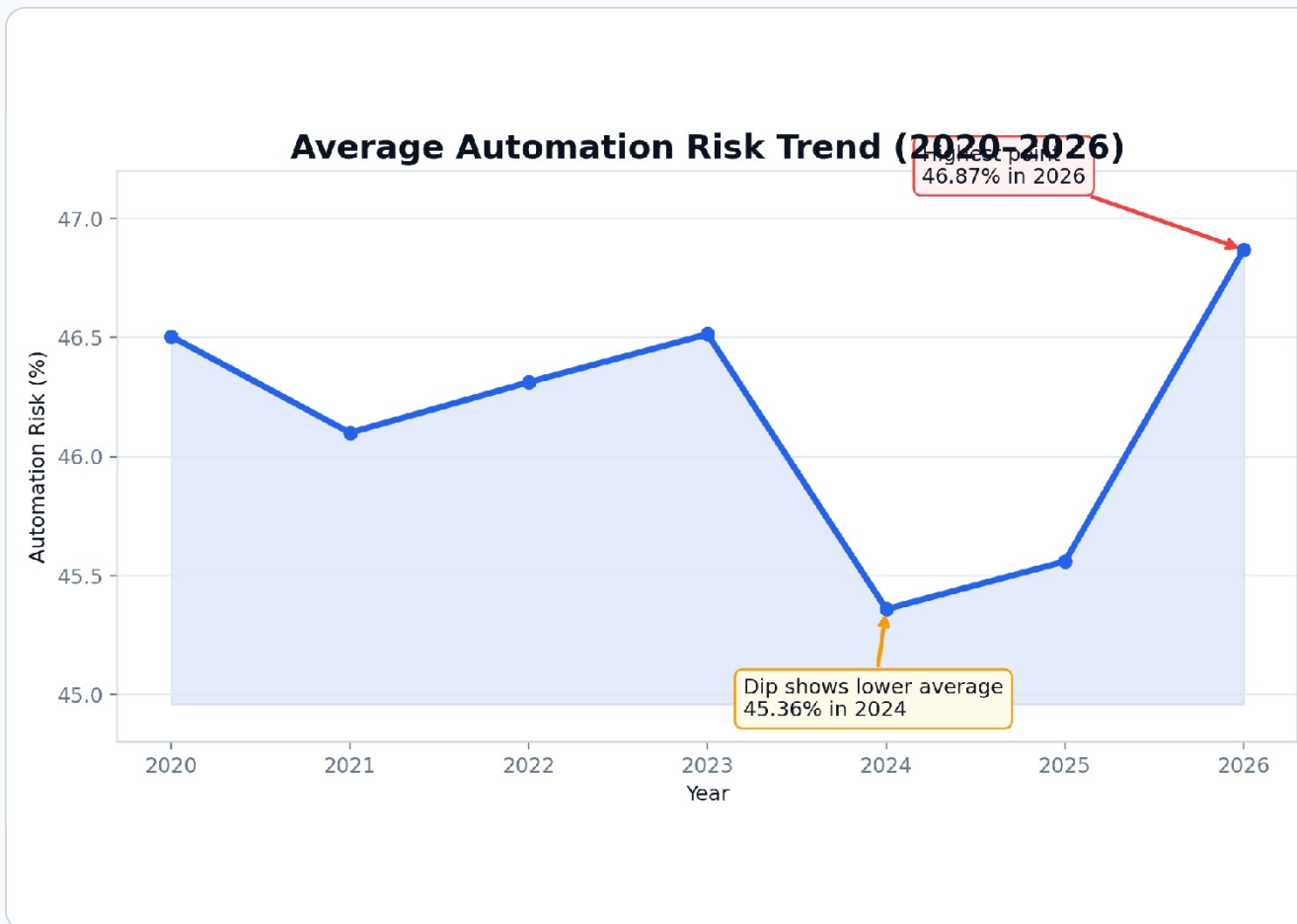
Why arrows are added

Pointers make the chart easier to read by highlighting spikes, dips, median values, and positive/negative zones.

Clean layout approach

Each slide has a clear title, one main visual idea, and short explanation so the teacher can understand it quickly.

Automation Risk Trend



Main insight

Average automation risk is around 46.18%. The chart shows yearly movement, not individual job changes.

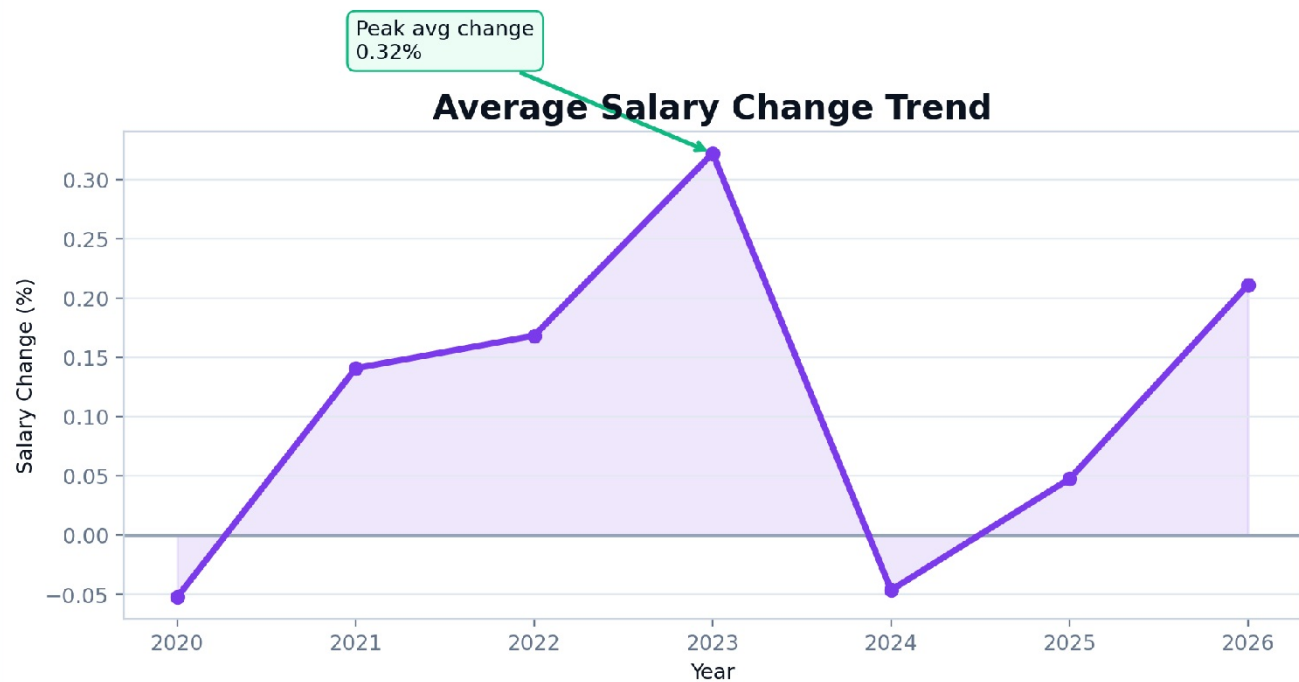
Why 2026 rises

2025 average risk $\approx 45.56\%$, while 2026 average risk $\approx 46.87\%$. Since the mean increased, the line moves upward.

Why 2024 dips

2023 average risk $\approx 46.51\%$, but 2024 average risk $\approx 45.36\%$. Lower average values create the downward dip.

Salary Change Trend



What this chart shows

It tracks average salary change percentage year by year. Positive values mean average salary increased after AI impact.

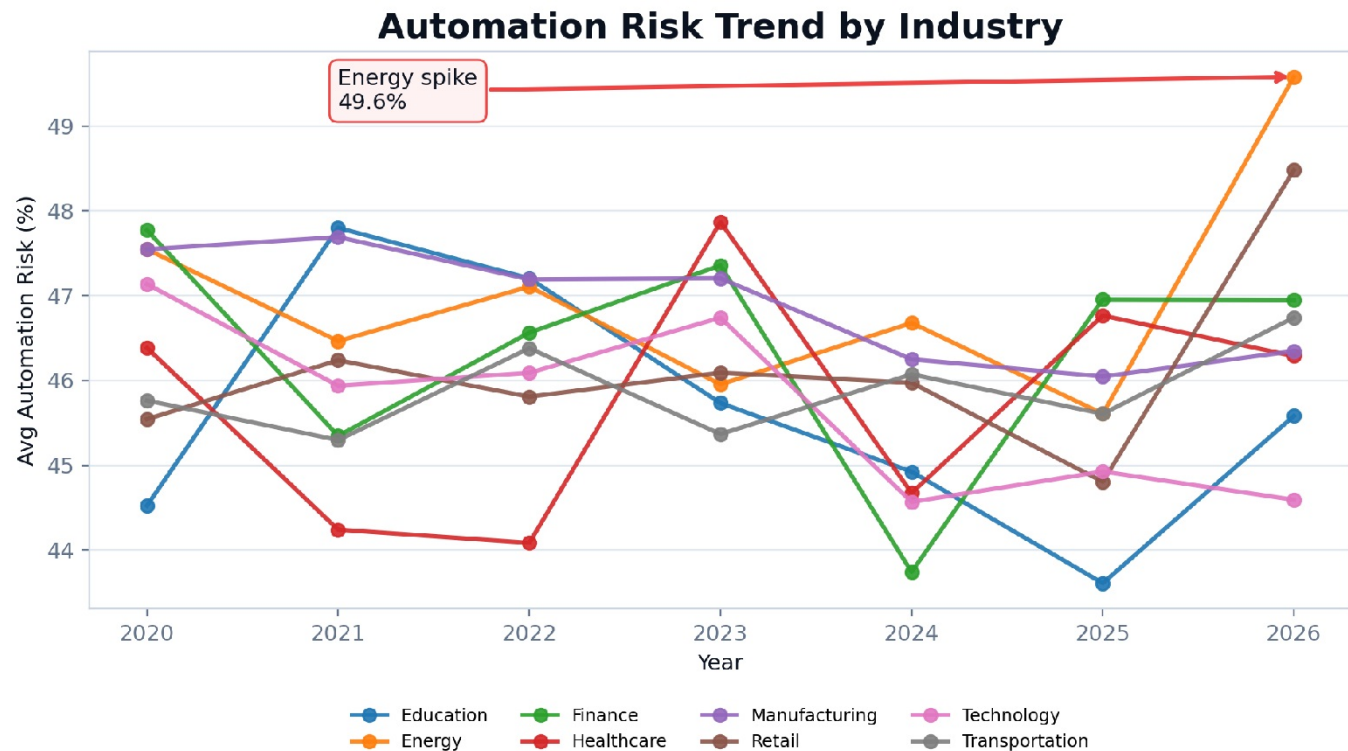
Interpretation

The average salary change is close to neutral (0.11%). This suggests AI impact is mixed: some roles gain value while some face pressure.

Simple reading rule

Line above 0 = salary growth. Line below 0 = salary decline. Higher peaks mean stronger average salary improvement.

Industry-Wise Automation Risk



Comparative view

Each line represents one industry. This makes it easy to see which sectors have spikes or stable risk.

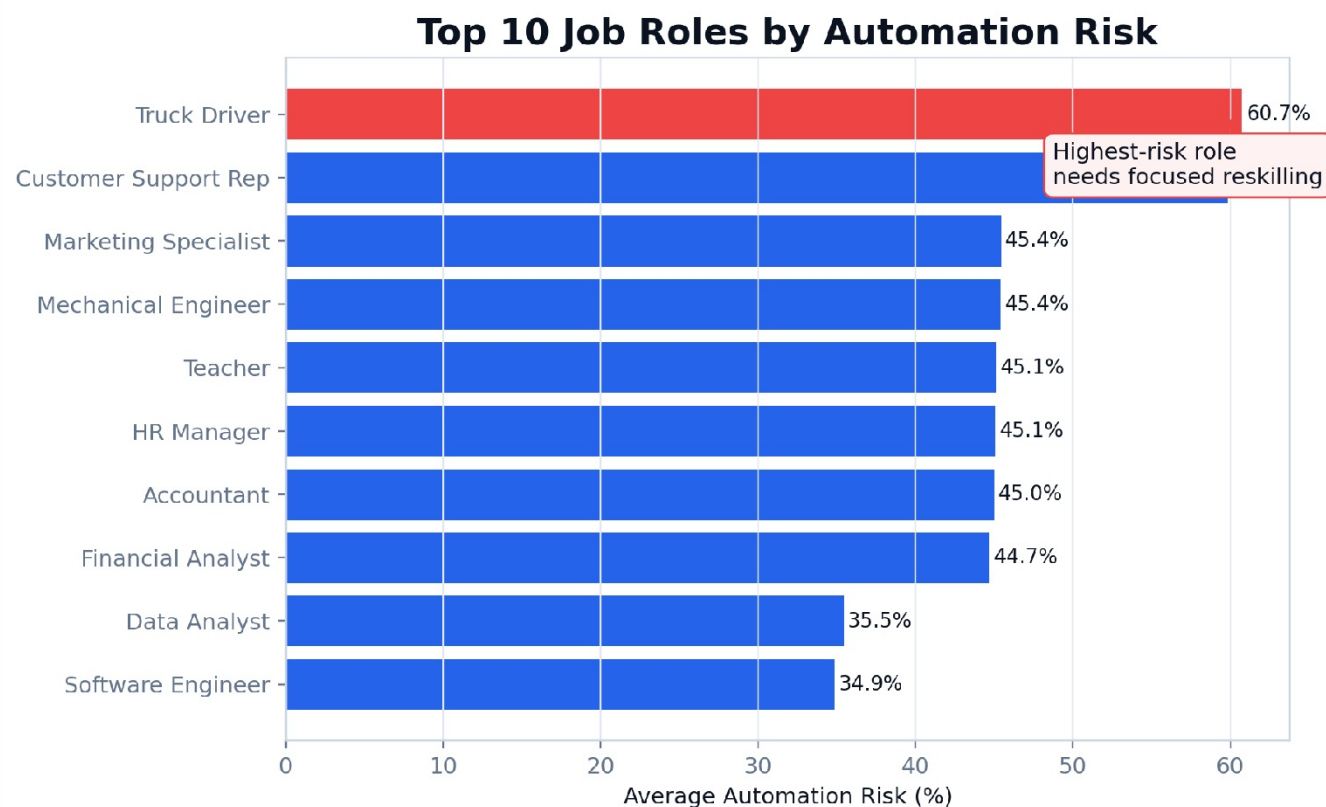
Uneven AI impact

AI does not affect all industries equally. Risk depends on task repetition, digital maturity, and adoption speed.

Best way to explain

A spike means that the average automation risk for that industry-year is higher than its previous year values.

Top Job Roles by Automation Risk



Highest risk role

Truck Driver has the highest average automation risk in this dataset, around 60.7%.

Highest-risk role needs focused reskilling

Why these jobs appear

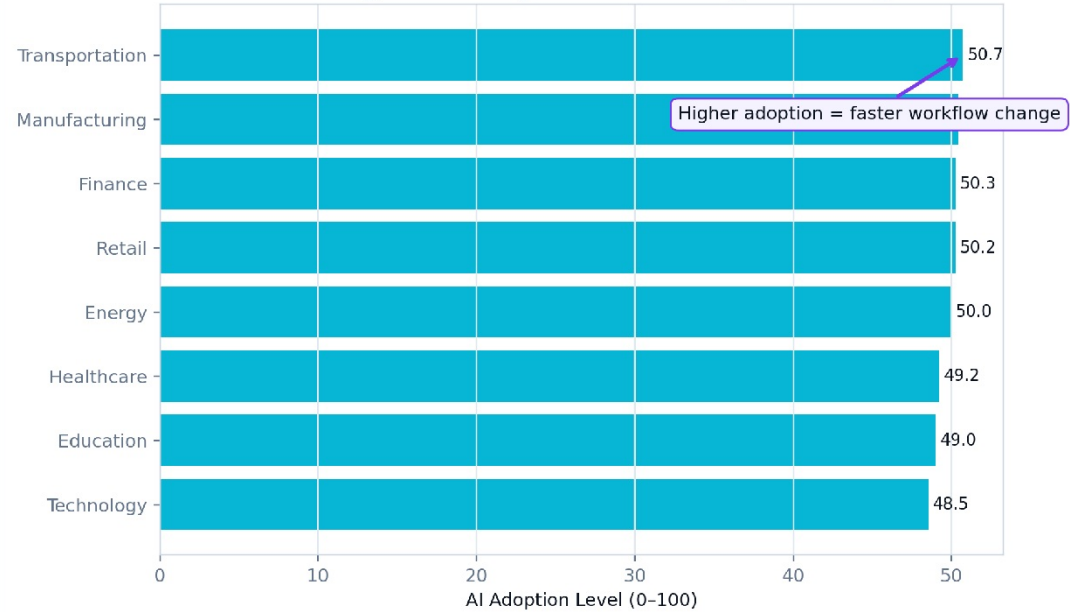
High-risk roles usually involve predictable, repetitive, measurable, or rule-based tasks that AI can support or automate.

Important takeaway

The chart highlights where reskilling should be prioritized. It does not mean full job loss; it means task transformation.

AI Adoption & Risk Distribution

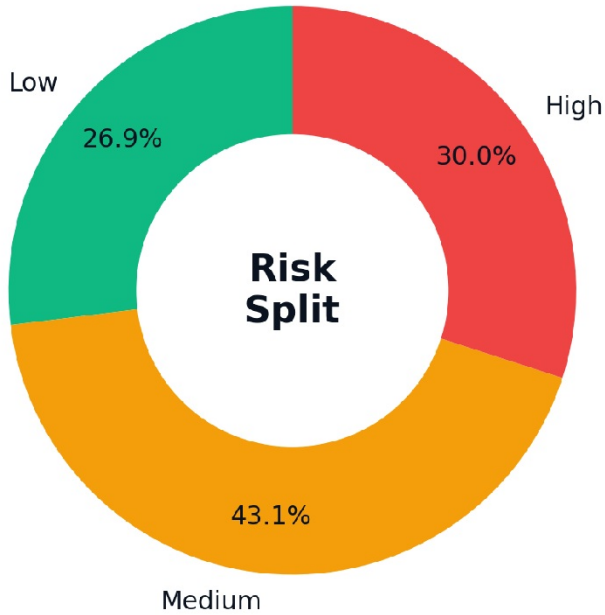
Average AI Adoption Level by Industry



Adoption view

Industry-level adoption shows how actively AI is entering workflows.

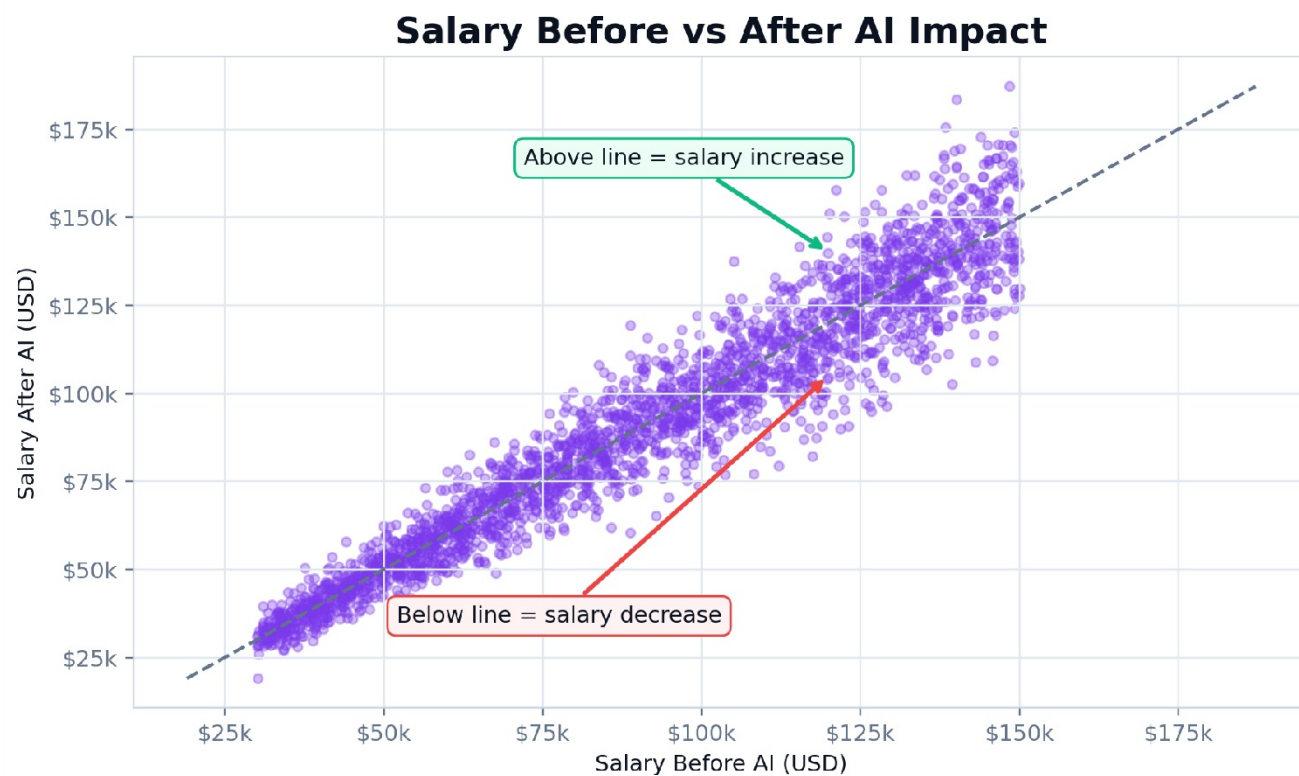
Automation Risk Category Distribution



Risk split

Risk categories summarize the overall vulnerability pattern.

Salary Before vs After AI



How to read it

Every point is one job record. X-axis is salary before AI; Y-axis is salary after AI.

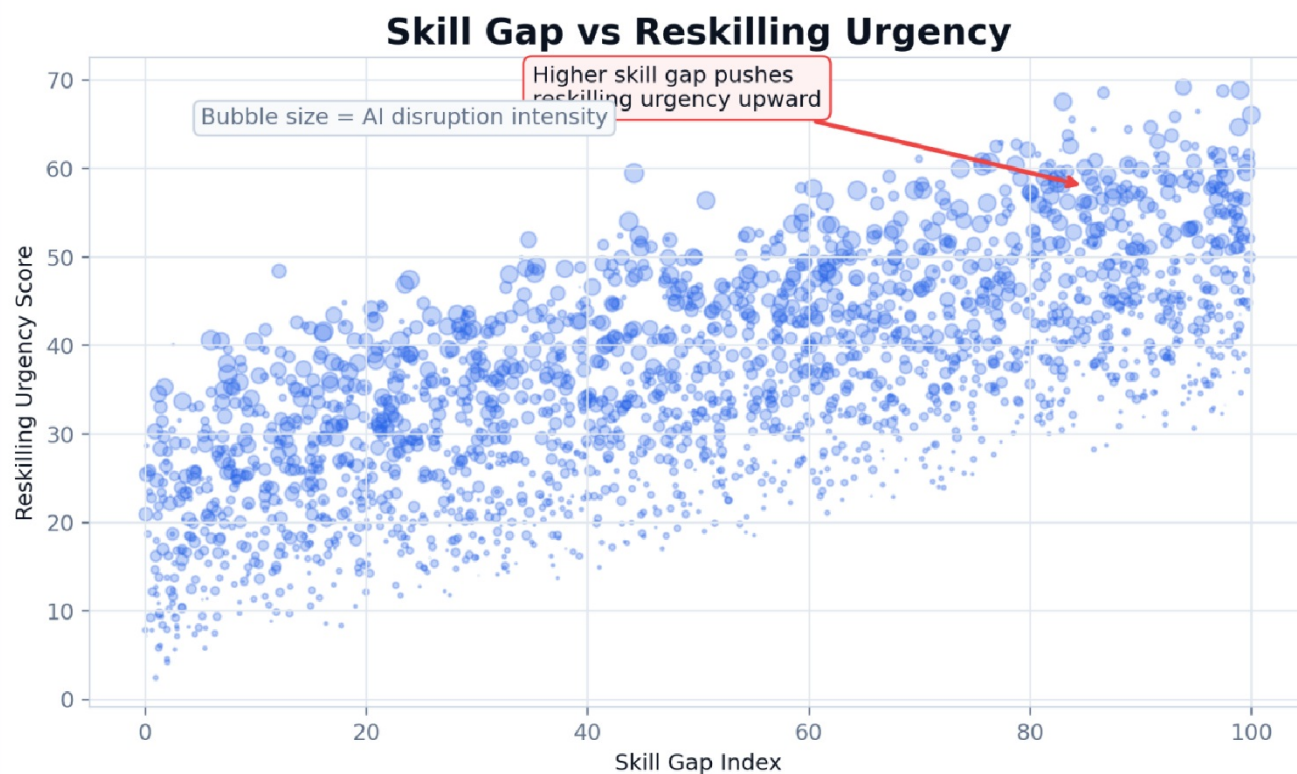
Reference line

The dotted line is the no-change reference. Points above it indicate salary increase; below it indicate decline.

Simple formula

Salary after AI = Salary before AI $\pm$ salary change.
That is why the points follow a strong upward pattern.

Skill Gap & Reskilling Urgency



What the bubbles mean

X-axis shows skill gap; Y-axis shows urgency to reskill. Larger bubbles mean stronger AI disruption.

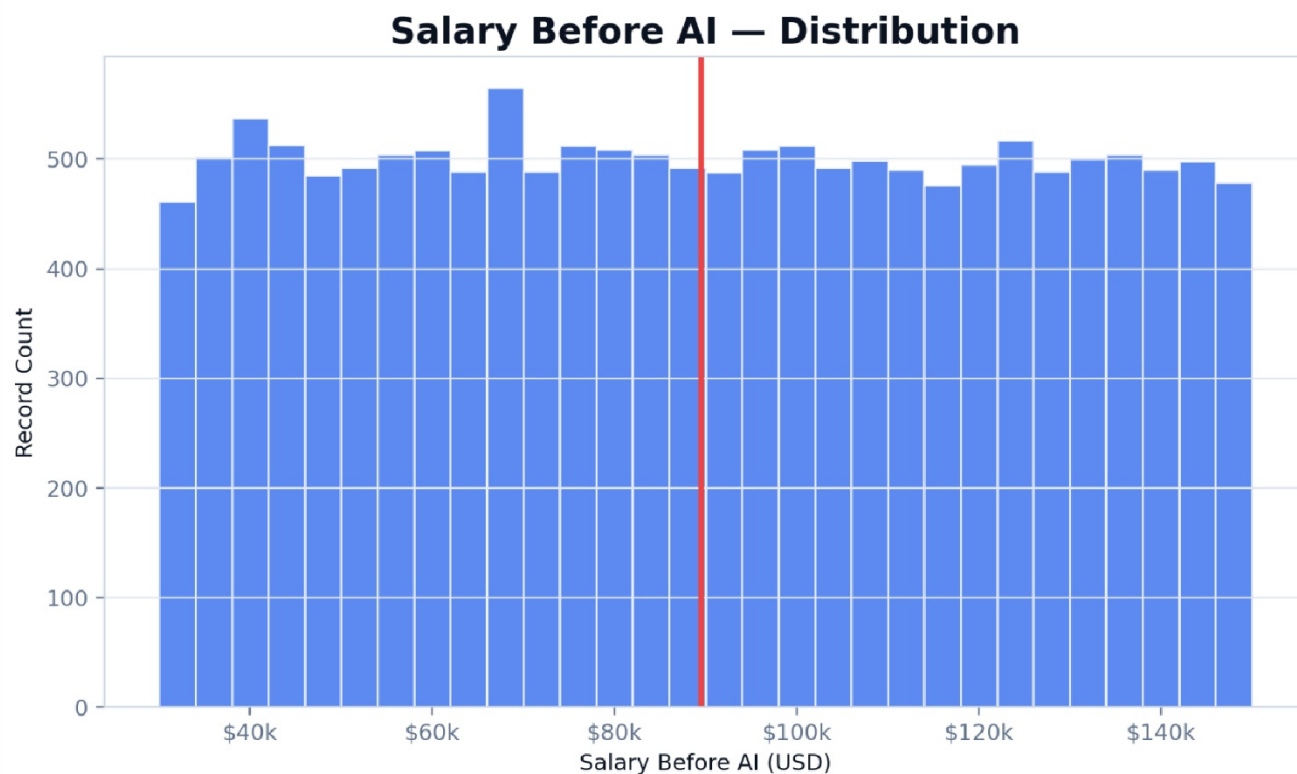
Main relationship

The upward pattern shows that higher skill gap generally leads to higher reskilling urgency.

Practical meaning

Workers in high-gap roles should focus on AI tools, analytical thinking, adaptability, and domain-specific upskilling.

Salary Distribution Before AI



Distribution view

Histogram groups salaries into ranges, showing where most observations are concentrated.

Median marker

The red line marks the middle salary level. Half the records are below this point and half are above it.

Why it matters

Salary distribution helps compare how AI impact may differ for lower, middle, and higher income roles.

Key Findings & Recommendations

1. Automation risk is significant

Average risk stays around the mid-40% range across 2020–2026, so AI impact is steady and visible.

2. Risk differs by role

Some job roles show noticeably higher automation risk because their tasks are easier to standardize or automate.

3. Salary effect is mixed

Average salary change is close to neutral, meaning AI creates both upward and downward pressure.

4. Skill gap drives urgency

The bubble chart shows that larger skill gaps usually require faster reskilling action.

5. Best response is upskilling

Learning AI tools, data skills, problem-solving, and domain expertise reduces replacement risk.

Conclusion: AI is not only replacing tasks; it is reshaping job roles. The strongest strategy is continuous reskilling and practical AI adoption.

References

Dataset

AI Job Replacement 2020–2026 dataset used for analysis. Variables include job role, industry, country, salary, automation risk, AI adoption, skill gap, and reskilling urgency.

R Libraries

ggplot2 for charts, dplyr for data grouping and summaries, tidyr for data preparation, corrrplot/GGally for extended analysis support.

Visualization Types

Line charts, horizontal bar chart, donut chart, histogram, scatter plot, and bubble chart were used to explain patterns clearly.

Thank You